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New paleomagnetic data from the Huanan Block, China, and Cretaceous tectonics in eastern China

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ABSTRACT

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A total of 72 grey–green fine-grained sandstone and siltstone core samples (from 10 sites) were collected from the Early Cretaceous Jieshan and Hekou Formations at Yongan, in the Huanan Block of southeastern China (Fujian Province). After removal of an erratic soft magnetic component in fields of 10–20 mT, stable magnetizations were retrieved from both rock units which are carried mainly by single-domain/pseudo-single-domain (SD/PSD) magnetite or titanomagnetite, as suggested by isothermal remanent magnetization, coercivity measurements and microscopic observations. The Hekou Formation yields a tilt-corrected pole at 66.9°N, 221.4°E ($n = 5$ sites, $K = 203.3$, $A_{95} = 5.4°$). A fold test on this result suggests that the magnetization was acquired before tilting which occurred shortly after deposition (between Early and Late Cretaceous). This magnetization is related to a primary sedimentary fabric, as indicated by the low-field anisotropy of magnetic susceptibility (AMS) measurements. The positive fold test and occurrence of a primary sedimentary fabric suggest a depositional origin for the stable magnetization. The Jieshan Formation gives a pole at 70.8°N, 228.7°E ($n = 5$ sites, $K = 47.7$, $A_{95} = 11.2°$) in geographic coordinates, which is statistically indistinguishable from the tilted pole for the Hekou Formation. The associated stable magnetization corresponds to a secondary fabric. These features indicate that the Jieshan Formation has been remagnetized during Cretaceous time. The predominance of SD/PSD magnetite or titanomagnetite and igneous activities in eastern Fujian during the Cretaceous suggest a viscous thermal remanent magnetization origin for the secondary magnetization.

The Early Cretaceous pole from the Hekou Formation is similar to those for the Yangzi Block, suggesting that the Huanan Block has been accreted to the Yangzi Block at least since Early Cretaceous time. A high consistency of Cretaceous poles from the Sino-Korean, Yangzi and Huanan Blocks as well as Hainan Island indicates that eastern China has been a coherent tectonic unit since Cretaceous time, although there might have been a small amount of northward displacement of the sampling area with respect to the Yangzi Block during the Early–Late Cretaceous period as a result of the overthrust of the Huanan Block onto the Yangzi carbonate platform. Comparison of Cretaceous poles for eastern China with the Eurasian apparent polar wander path (APWP) suggests that no paleomagnetically detectable relative motion has taken place since the Cretaceous. The Cretaceous poles of eastern China, however, display a paleomagnetically significant westward motion (about 9° in angular distance), which is not observed in the Eurasian APWP. This difference may be indicative of a younger age corresponding to the Cretaceous–Tertiary boundary for all Late Cretaceous poles from eastern China, or more probably may imply a minor error in the definition of the Eurasian APWP.

1. Introduction

The geology of South China has been interpreted as a post-Caledonian (Early Paleozoic) platform (Huang, 1977, 1980; Zhang et al., 1984; Yang et al., 1986). According to this notion, South China is divided into the Yangzi Craton to the northwest and the South China Fold System to the southeast. The Yangzi Craton was believed to be of Proterozoic age (700–800 Ma). The South China Fold System was regarded as a tectonic

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unit which formed during the Caledonian tectonic movement and which remained a platform since then. Late in the Mesozoic, this 'platform' was strongly folded, overthrust and invaded by extensive granitic magmatism and acidic volcanism. These Mesozoic activities have been interpreted as a result of subduction of the Pacific Plate beneath mainland Asia (Hamilton, 1979; Huang, 1984; Ren, 1984; Wang, 1986). Consequently, the term 'paraplatform' was introduced to designate this deformed system, to distinguish it from the notion of 'platform' (Huang, 1977, 1980).

This old notion has faced challenges after the emergence of new geological data in the 1980s. In 1981, Hsü pointed out that the notion of a 'South China Platform' is erroneous and misleading, owing to the evidence of extensive Mesozoic deformation of the 'platform sediments' and coeval convergent-margin magmatic activities. He also suggested that the tectonic pattern of South China is largely comparable with that of the Central and

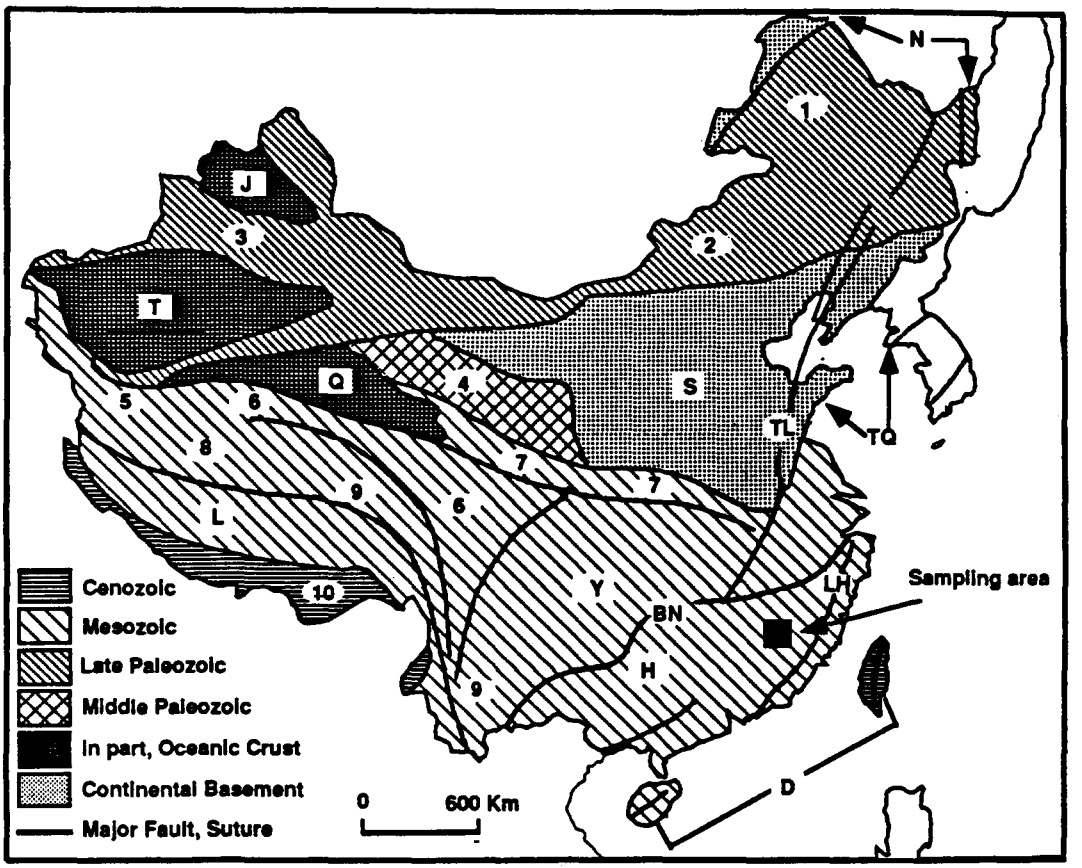


Fig. 1. Generalized tectonic map of China (simplified from Hsü et al. (1990)). Major blocks: N, Northeast China Block; J, Jungar Block; T, Tarim Block; Q, Qaidam Block; S, Sino-Korean Block; Y, Yangzi Block; H, Huanan Block; D, Dongnanya Block. Major orogenic belts: 1, Xinganling Mts.; 2, Yinshan Mts.; 3, Tianshan Mts.; 4, Qilian Mts.; 5, Kunlun Mts.; 6, Songpan–Gantze Deformed Belt; 7, Qinling–Dabie Mts.; 8, Tanggula Deformed Belt; 9, Sanjiang Deformed Belt; 10, Himalaya Mts. Major faults and sutures associated with the present study: TL, Tancheng–Lujiang Fault zone; TQ, Tanlu–Qingdao Suture zone; BN, Banxi–Nanpanjiang Suture zone; LH, Lishui–Haifeng Fault zone.

Southern Appalachians. As a result, South China may be regarded as an orogenic belt of the collisional type rather than a post-Caledonian platform. Hsü's model was largely supported by subsequent studies (Sengör, 1985; Sengör et al., 1988; Hsü et al., 1988a, b, 1989a, b; Li et al., 1989). In a recent study by Hsü et al. (1990), South China was divided into the Yangzi, Huanan (equivalent to Huang's (1977) South China Fold System in areal extent) and Dongnanya (or Southeast Asia) Blocks, separated by the Banxi–Nanpanjiang and Gunanhai Suture zones (Fig. 1). According to this model, the Huanan Block was separated from the Yangzi Block to the north by a Banxi–Nanpanjiang Ocean of Late Precambrian–Early Mesozoic age and from the Dongnanya Block to the south by a Gunanhai Ocean of Late Paleozoic–Early Mesozoic age. The Huanan Block collided with the Yangzi Block during the Middle Triassic, and with the Dongnanya Block during the Triassic or Jurassic.

In the light of Hsü's model, the traditional view of a 'South China Platform' deserves further testing. Paleomagnetism can be used to constrain the geological history of blocks and is therefore important in testing the validity of this new model. Unfortunately, only a few Cretaceous paleomagnetic results have been reported from this block (Li et al., 1982; Hu, 1986; Chan and Zich, 1986; Hsü, 1987; Lee et al., 1987; Chan, 1991), because of the complex regional geology, generally weak intensity of magnetization for these rocks and intense Late Mesozoic–Cenozoic remagnetization. This greatly restricts the test of validity of Hsü's model using paleomagnetism. It was thus decided to conduct a paleomagnetic sampling during summer 1989 at Yongan and Longyan, in Fujian Province, at the southeastern rim of the Huanan Block. This paper presents a new paleomagnetic pole, determined from Early Cretaceous sedimentary rocks in the Yongan area (see Fig. 1), and represents a part of our investigation. The Cretaceous tectonics in eastern China is also discussed in combination with other Cretaceous poles from the Sino-Korean, Yangzi, Huanan and Dongnanya Blocks and through their comparison with the Eurasian apparent polar wander path (APWP) defined by Besse and Courtillot (1991).

2. Regional geology

According to the regional geology (Fujian Bureau of Geology, 1985), Fujian may be divided into three tectonic domains: the East Fujian volcanic fault–fold belt, the Southwest Fujian fault-subsided belt and the Northwest Fujian folded rise (Fig. 2). The East Fujian volcanic fault–fold belt is separated from the Southwest Fujian fault-subsided belt by the Zhenghe–Dapu deep fault zone. This fault is regarded as part of the very extensive NE–SW-trending Lishui–Haifeng fault system in Fujian Province and plays an important role in the geological development of the eastern and western parts of this fault.

The Yongan area is located in the Southwest Fujian fault-subsided belt, where the Upper

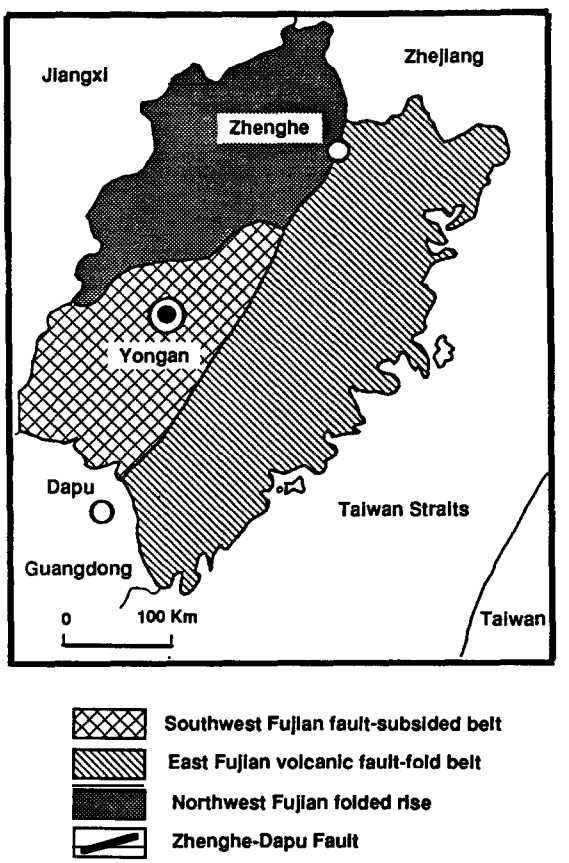


Fig. 2. Tectonic subdivisions of Fujian province, showing the sampling locations.

Mesozoic sequences are well developed. Unmetamorphosed Lower Cretaceous sediments consist of the Bantou and Jieshan Formations (lacustrine–estuarine facies) in the lower–middle section, and the Hekou Formation (continental facies) in the upper section. The Bantou Formation consists of grey–green or grey–black tuffaceous sandstones. The Jieshan Formation is dominated by grey–green, grey–black fine-grained sandstones, siltstones and paper shales. The Hekou Formation comprises purplish red conglomerates, coarse-grained sandstones in the lower part and grey–green calcareous fine-grained sandstones and siltstones in the upper part. Stratigraphical and paleontological studies have indicated that the Bantou and Jieshan Formations were deposited during the early stage of the Early Cretaceous, and the Hekou Formation during the middle–late stage of the Early Cretaceous (Aptian–Albian period) (Fujian Bureau of Geology, 1985).

Since the beginning of the Cretaceous, the region has been affected mostly by the late stages of the Yanshanian orogenic activity (i.e. from the beginning of the Early Cretaceous to the end of the Late Cretaceous). Two major phases of tectonic activity are identified. The first phase took place between the Early and Late Cretaceous. It is recorded by an unconformity between the Lower Cretaceous Hekou Formation and the Upper Cretaceous Shaxian Formation composed of continental redbeds. This event is characterized by a large-scale uplift to the east of the Zhenghe–Dapu fault and the formation of subsidence basins to the west. In general, folding is weak and is dominated by open folds or monoclinal structures. Granitic intrusions, however, were very active, particularly to the east of the Zhenghe–Dapu fault zone, which marks the most intense magmatic activity in the region since Cretaceous time. The more recent phase occurred at the end of the Cretaceous. It is recorded by a disconformity or unconformity between the Late Cretaceous Chishi Group composed of continental red coarse-grained clastic rocks (or other underlying beds) and the Upper Tertiary Futang Group of littoral-embayment facies. The absence of Lower Tertiary rock units suggests a large-scale uplift during this tectonic activity (Fujian Bureau of Geology, 1985). The present tectonic framework of this region was finally formed after this phase of tectonic activity.

3. Paleomagnetic sampling and laboratory measurements

Two localities around Yongan City were chosen for the sampling: Jieshan (26°N, 117.4°E) and Putou Village (26°N, 117.2°E). At Jieshan, 37 (sites 1–5) grey–green fine-grained sandstone and siltstone samples were collected from the Jieshan Formation; the thickness of the formation covered by the sites is only about 80 m. In Putou Village, 35 samples (sites 28–32) were taken from grey–green or grey–black calcareous siltstones and sandstones in the upper part of the Hekou Formation; the thickness of the formation covered by these sites is roughly 500 m.

Samples were taken using portable drilling equipment, and were oriented with a magnetic compass. At least six cores were drilled per site. The bedding attitudes were determined as an average of at least three independent measurements which were taken from localities with unequivocal parallel bedding orientations. The sites at Jieshan display uniform bedding. The sites at Putou Village, however, indicate some changes in the bedding attitude (315–330° for the strike and 19–35° for the dip), which allowed performance of a fold test at this locality.

Magnetic measurements and demagnetizations were conducted at Stanford University using an automated cryogenic magnetometer with integrated alternating field (a.f.) demagnetization coils. Generally only one specimen from each sample was used for paleomagnetic determination. The specimens were a.f. demagnetized over 8–16 steps up to 80 mT. Components of magnetization isolated during demagnetization were identified using standard orthogonal demagnetization plots (Zijderveld, 1967) and their directions were determined using principal component analysis (Kirschvink, 1980) with maximum angular deviations smaller than 10°. Site means and unit means were calculated using the method of Fisher (1953).

Before the process of demagnetization, low-field anisotropy of magnetic susceptibility (AMS) of each specimen was measured with a Sapphire Instruments (Ruthven, Ont., Canada) SI-2 induction coil instrument coupled with an IBM personal computer, in an attempt to investigate the origin of remanent magnetizations. To identify magnetic minerals and their domain structures in the rocks investigated, some typical sister specimens were subjected to acquisition and destruction of isothermal remanent magnetization (IRM) in field up to 800 mT using a home-made static magnet, to microscopic observations and to thermal demagnetization using a CTF (Port Coquirtlam, B.C., Canada) cryogenic magnetometer and a Schonstedt (Reston, VA) TSD-1 oven at Windsor University.

4. Paleomagnetic results

Table 1 summarizes the paleomagnetic data from the Jieshan and Hekou Formations. Characteristic magnetizations with a single normal polarity were isolated at all sites. The directions within and between sites for each rock unit are generally consistent. Examples of orthogonal demagnetization plots for specimens from both formations are shown in Figs. 3 and 5 respectively.

4.1. Hekou Formation (sites 28–32)

For most specimens, the natural remanent magnetization (NRM) comprises two components, as indicated by specimens 30-7, 31-5 and 32-1 in Fig. 3. The soft component is characterized by a low coercivity (10–20 mT) and displays erratic directions, suggesting that it is of recent viscous origin. The stable component has a normal polarity and is isolated in the 20–80 mT coercivity range. This stable component is interpreted as the characteristic magnetization for the Hekou Formation on the basis of linear trajectories of demagnetization towards the origin and a similar direction from specimen to specimen. A single component of NRM is found in some specimens, as shown by specimen 29-2 in Fig. 3, in which the whole demagnetization trajectory is nearly linear towards the origin until the intensity drops near noise level at 80 mT. This single component is compatible with the stable component that was found to be superimposed by a soft viscous component (in direction and coercivity);

TABLE 1

Late Mesozoic paleomagnetic data from Yongan, Fujian

Sites no.	N/n	In situ paleomagnetic data						Tilted paleomagnetic data					
		D_g	I_g	$\kappa\,(K)$	$\alpha_{95}\,(A_{95})$	λ_p	ϕ_p	D_s	I_s	$\kappa\,(K)$	$\alpha_{95}\,(A_{95})$	λ_p	ϕ_p
1	7/7	23.4	42.3	136.2	5.2	68.8	206.4	21.2	−7.3	135.9	5.2	53.9	259.6
2	8/8	17.6	33.8	21.7	12.2	72.0	228.5	16.2	−5.8	21.7	12.2	57.1	266.5
3	8/7	2.1	20.2	50.5	8.6	74.3	289.6	1.6	−24.4	50.8	8.6	51.2	294.9
4	7/6	19.0	34.7	22.0	14.6	71.2	225.0	19.1	−15.3	21.9	14.6	51.4	266.0
5	7/5	32.4	28.9	42.1	11.9	58.1	220.2	31.4	−20.0	42.1	11.9	42.6	253.3
Unit mean		18.5	32.4	41.4	12.0			18.0	−4.8	39.3	12.4		
(sites 1–5)				(47.7)	(11.2)	70.8	228.7			(50.7)	(10.8)	52.1	267.5
28	6/5	19.8	52.9	28.5	14.6	71.3	179.1	26.7	35.2	28.6	14.5	64.5	216.8
29	6/6	337.7	44.6	62.6	8.5	70.0	33.1	25.7	37.8	63.2	8.5	66.0	213.5
30	8/6	0.3	53.6	29.2	12.6	81.9	119.0	19.5	26.9	29.1	12.6	68.3	236.5
31	7/6	8.0	46.4	94.5	6.9	82.6	192.1	20.7	25.3	94.8	6.9	66.8	236.6
32	8/4	3.1	60.3	139.3	7.8	74.6	126.2	25.9	42.2	137.9	7.8	66.6	205.6
Unit mean		1.1	52.4	48.2	11.1			23.5	33.5	109.8	7.3		
(sites (28–32)				(31.7)	(13.8)	82.5	127.5			(203.3)	(5.4)	66.9	221.4

$N(n)$, number of individual specimens measured (accepted) at a site; D_g (D_s) and I_g (I_s), site mean directions before (after) tilt corrections; $\kappa\,(K)$, precision parameters of site mean directions (unit mean poles); α_{95} (A_{95}), 95% confidence circles of site mean or unit mean directions (unit mean poles); λ_p and ϕ_p, latitudes (°N) and longitudes (°E) of paleomagnetic poles.

consequently, it can be regarded as the characteristic magnetization.

Figure 4 shows an equal-area projection of site mean directions before and after tilt corrections. Before tilt correction, the characteristic magnetization displays a northward direction with a steep downward inclination (D (declination) = 1.1°, I (inclination) = 52.4°, κ (precision parameter of site mean direction) = 48.2, α_{95} (95% confidence circle of site mean direction) = 11.1°); after tilt correction, it is oriented northeast with an intermediate downward inclination (D = 23.5°, I = 33.5°, κ = 109.8, α_{95} = 7.3°) It is noted that the present Earth's field (PEF) direction (D = 2°, I = 43°) of the locality falls into the 95% confidence circle of the unit mean in situ direction (Fig. 4(a)). However, the steeper inclination at sites 28, 30 and 31 than for the PEF, the pronounced

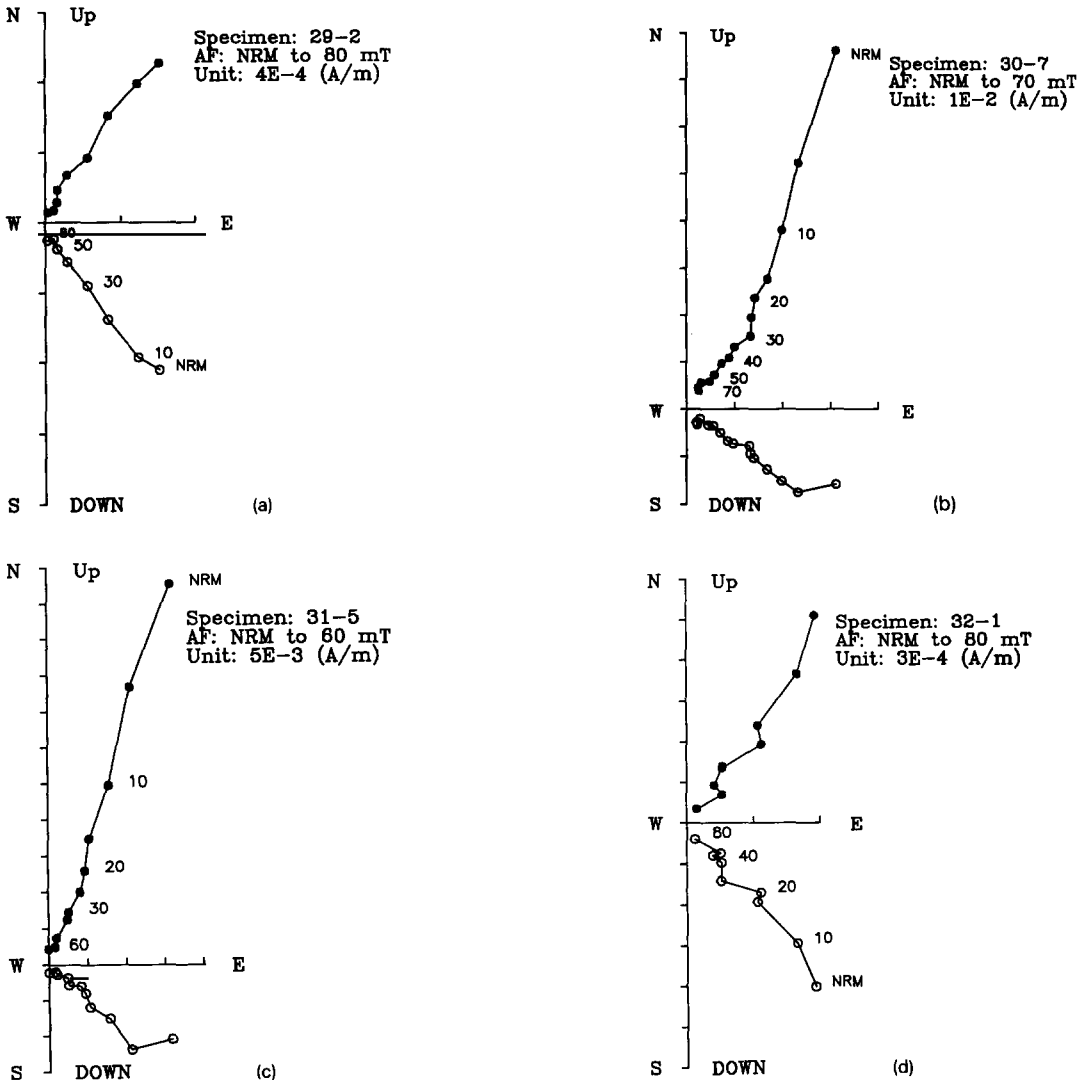


Fig. 3. Orthogonal a.f. demagnetization diagrams of four typical specimens from the Hekou Formation (after tilt correction). Solid (open) circles represent projections on the horizonal (vertical) plane. Numbers next to the circles are demagnetization steps (mT).

deviation of declination from that of the PEF at sites 28 and 29 and the fact that the concentration of site mean directions is much improved after tilt correction ($\kappa_a/\kappa_b = 2.2780$) precludes the possibility of a PEF overprinting. Although the characteristic direction does not pass the fold test statistically by means of McElhinny's method (1964), because of insufficient variations of bedding attitude, the pronounced improvement in the grouping of site mean directions after the tilt correction and a positive fold test for paleomagnetic poles at 99% confidence level ($K_a/K_b = 6.4132$ (where K is the precision parameter of virtual geomagnetic poles)) suggest that the characteristic remanence was acquired before the second phase of tectonic activity (pre-Late Cretaceous times). The normal polarity at all sites corresponds to the Cretaceous normal superchron, which is in agreement with the age of the formation and with the result of the fold test.

4.2. *Jieshan Formation (sites 1–5)*

A single and stable component of NRM is a predominant magnetic signature of this formation, as shown by specimens 1-2, 2-1 and 3-8 in Fig. 5. Combinations of soft and stable components were observed in some specimens. Specimen 4-5 shows a typical example of this behaviour. The soft component, when present, is usually found in the low coercivity range (10–20 mT) and displays erratic directions from specimen to specimen. This behaviour is exactly the same as that observed in the Hekou Formation, and, therefore, the soft component is of recent viscous origin. The stable component, whether isolated after removal of the soft component or caused by a single component, usually shows a similar direction and is identified up to 80 mT until the linear trajectories of demagnetization reach the origin of the orthogonal vector diagram. It is therefore interpreted as the characteristic magnetization for this rock unit.

It can be seen from Fig. 5(b) that the characteristic component in geographic coordinates displays a northeasterly directed magnetization with intermediate downward inclination ($D = 18.5°$, $I = 32.4°$, $\kappa = 41.4$, $\alpha_{95} = 12.0°$). This direction is statistically distinguishable from the PEF direction at this locality, indicating that it is not due to PEF overprinting. Upon tilt correction, however, it remains in the northeast direction with a shallow upward inclination ($D = 18.0°$, $I = -4.8°$, $\kappa = 39.3$, $\alpha_{95} = 12.4°$). Although uniformity of bedding attitudes at these sites prevents further constraint of the age of acquisition of the characteristic component, the possibility of early Cretaceous overprinting cannot be ruled out because of the resemblance of the characteristic direction in geographic coordinates to that of the pre-folding remanence in the Hekou Formation. Further comments on this issue follow below.

4.3. *Magnetic mineralogy*

Magnetic hysteresis as well as thermal and a.f. decay properties as follows were used as an aid to diagnose magnetic minerals and their domain structures: (1) isothermal remanence saturation magnetization J_{sr}; (2) remanent coercivity force

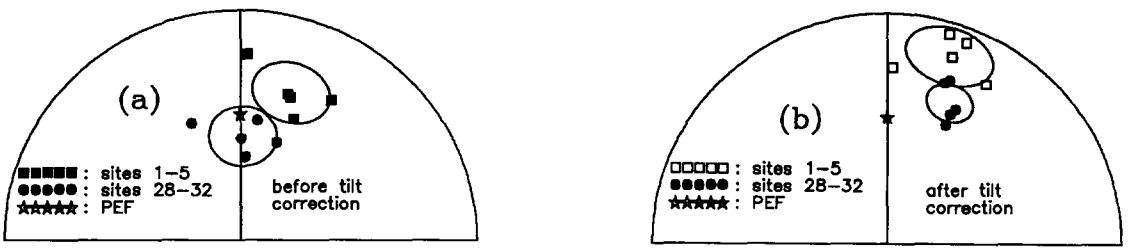


Fig. 4. Equal-area projections of site-mean directions before (a) and after (b) tilt correction. Solid (open) symbols are projections on the lower (upper) hemisphere. The star represents the present Earth's field (PEF), and 95% confidence angles (α_{95}) for the unit mean directions are plotted as small circles.

H_{cr}; (3) acquisition remanent coercivity force H'_{cr}; (4) median destructive field $H_{1/2I}$ of IRM; (5) ratios H'_{cr}/H_{cr} and $H_{1/2I}/H_{cr}$; (6) unblocking temperature T_{UB}, (7) median destructive field $H_{1/2N}$ of NRM.

Figure 6 shows IRM curves of six specimens used in Figs. 3 and 5, indicating the orthogonal vector diagrams. The IRM acquisition curves in all specimens reached saturation at fields between 100 and 150 mT, and the corresponding reverse field IRM curves yield low values of H_{cr} (approximately 30 mT), which would be typical of magnetite (titanomagnetite) and/or pyrrhotite. Magnetite (titanomagnetite) and pyrrhotite can be distinguished by investigating unblocking temperature in the thermal decay curve. As can be

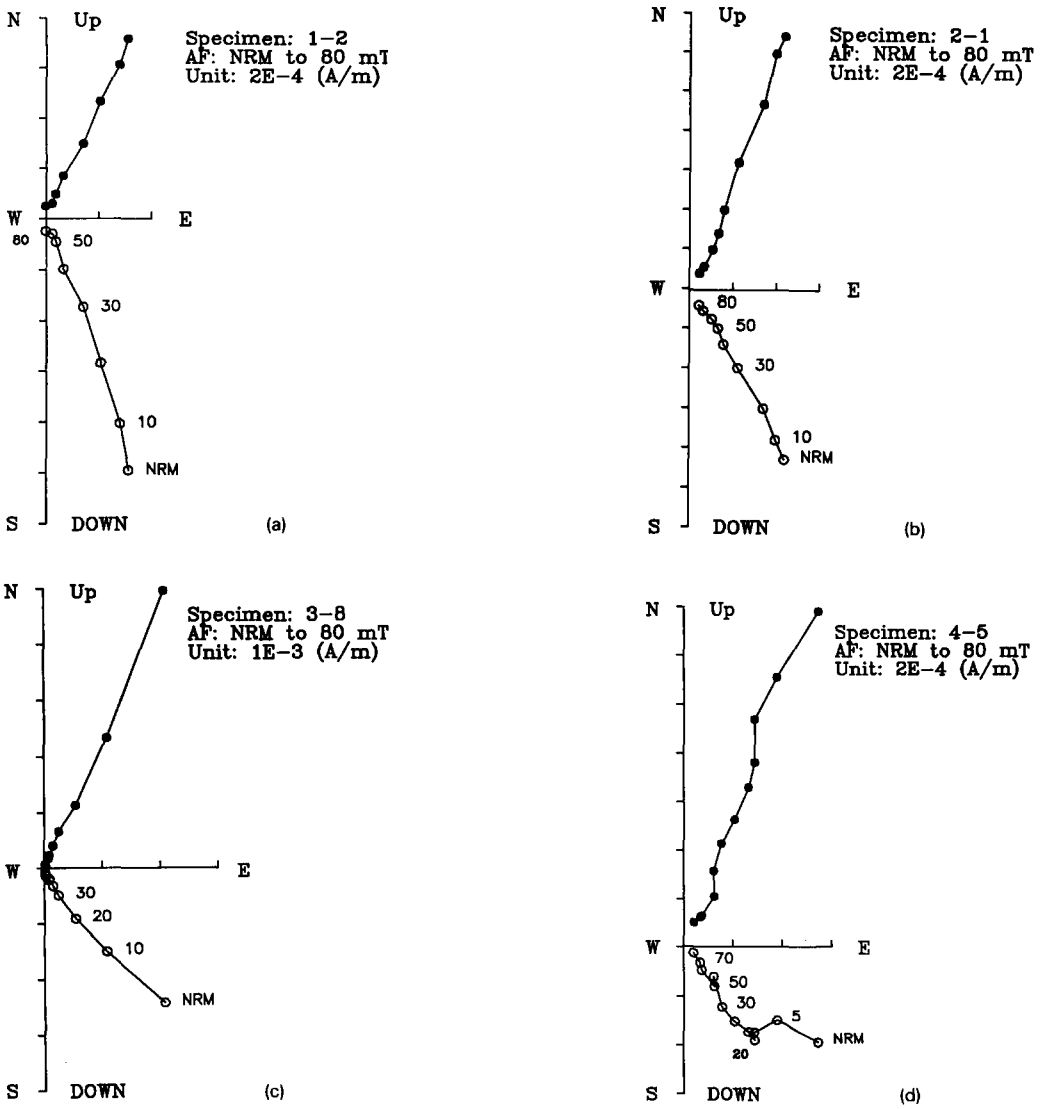


Fig. 5. Orthogonal a.f. demagnetization diagrams of four typical specimens from the Jieshan Formation (before tilt correction). Solid (open) circles represent projections on the horizonal (vertical) plane. Numbers next to the circles are demagnetization steps (mT).

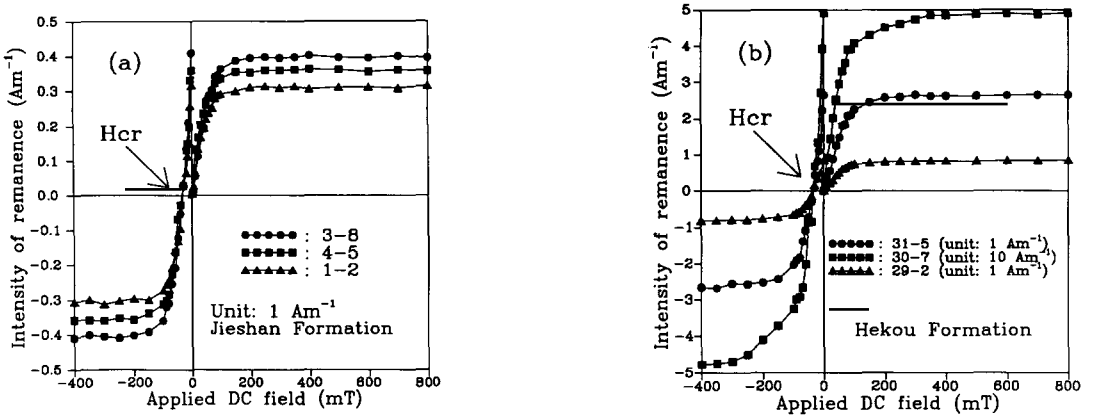


Fig. 6. IRM acquisition and destruction curves of typical specimens from (a) the Jieshan Formation and (b) the Hekou Formation, showing that titanomagnetite is the dominant magnetic carrier in the rocks investigated. (Note that these specimens are the same as those used in Figs. 3 and 5.)

seen in Fig. 7, the unblocking temperature spectra in all four specimens are typically distributed in the 400–600°C range. This behaviour could not be that of pyrrhotite, for which the remanence would rapidly decrease between 250 and 325°C (Dekkers, 1989). As the unblocking temperature occurs over a wide range instead of in the proximity of the Curie temperature of magnetite (575°C), titanomagnetite with varying Ti content is regarded as the principal ferrimagnetic mineral in both rock units.

Figure 8 illustrates a.f. decay curves for specimens used in Figs. 3 and 5. The median destructive field of NRM ($H_{1/2N}$) of all the specimens is located in the 10–30 mT range, which falls into the $H_{1/2N}$ range for single-domain type (SD) or bimodal type, as discussed by Dunlop (1983). A close inspection of these curves indicates that below an a.f. of 20 mT, the curves in some specimens (3-8, 4-5, 30-7, 31-5 and 32-1) have a multidomain (MD) configuration (as indicated by nearly linear decay curves), whereas above 20 mT they invert to an SD/PSD (pseudo-single-domain) configuration (as indicated by the inflected decay curves). Consequently, the domain structure in these specimens corresponds to the bimodal type described by Dunlop (1983) and Lowrie and Fuller (1971). However, an SD/PSD-type configuration is more likely to exist in specimens 1-2, 2-1 and 29-2, as indicated by the relatively upward inflected a.f. decay curves.

Table 2 gives values of hysteresis parameters and ratios of these parameters determined from the IRM curves. If these values are compared with their counterparts obtained by Hartstra (1982), two significant points can be made: (1) these values are much closer to those of Hartstra's DKA (optically homogeneous titanomagnetite) or LM6 (titanomagnetite showing exsolution lamellae) than those of HM4 (optically homogeneous magnetite) or HM7 (maghemitized titanomagnetite with fine-grained ilmenite exsolution); (2) the grain sizes approximately correspond to a

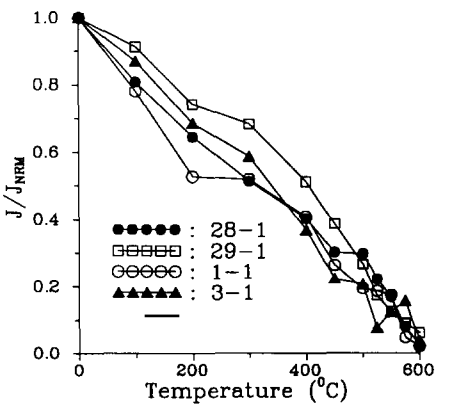


Fig. 7. Normalized thermal demagnetization curves of four typical specimens from both the Hekou (28-1 and 29-1) and Jieshan (1-1 and 3-1) Formations, showing the ranges of unblocking temperatures.

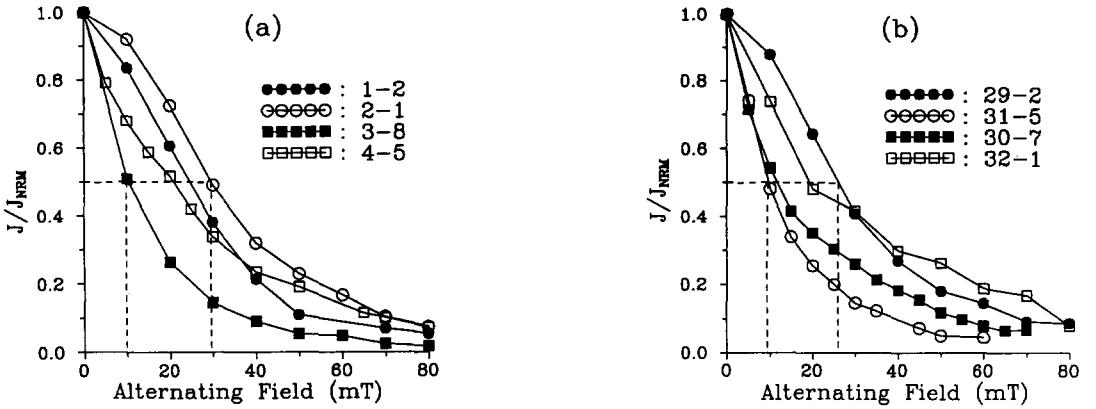


Fig. 8. Normalized a.f. demagnetization curves of typical specimens from (a) the Jieshan Formation and (b) the Hekou Formation. Intervals between the intersections of vertical dotted lines and the x-axis show the ranges of median destructive field $H_{1/2N}$ of NRM.

range of less than 15 μm in the case of DKA and 10–55 μm in the case of LM6. Therefore, this comparison confirms that titanomagnetite is the main magnetic carrier and its domain structure is of bimodal type, i.e. mixed SD/PSD (grain size less than or equal to 4 μm/4–15 μm; e.g. Dunlop, 1983) and MD (greater than or equal to 15 μm; e.g. Dunlop, 1983) types.

In summary, both the Jieshan and Hekou Formations have similar ferrimagnetic mineralogy which is dominated by titanomagnetite with a bimodal configuration. Large MD titanomagnetite grains are responsible for the recent viscous component, and SD/PSD titanomagnetite grains are more possibly the main carrier of the characteristic components isolated from the two rock units.

5. Magnetic fabric and origin of the characteristic magnetizations

Magnetic anisotropy is based on the total magnetic mineral content rather than the remanence that is carried by only a small fraction of the magnetic minerals; therefore, it is a very sensitive indicator of rock alterations (Hrouda, 1982; Jeleńska and Hofmokl, 1990). In undeformed sedimentary rocks, for example, the sedimentary fabric is mostly characterized by low degrees of

TABLE 2

Rock magnetic parameters and their ratios

Specimen	J_{sr}	H_{cr}	H'_{cr}	$H_{1/2I}$	H'_{cr}/H_{cr}	$H_{1/2I}/H_{cr}$
	(A m^{-1})	(mT)	(mT)	(mT)		
3-5-2	0.40	35	45	13	1.2857	0.3714
4-2-2	0.36	28	30	15	1.0710	0.5357
5-6-2	0.32	23	30	12	1.3043	0.5217
30-5-2	27.0	35	45	14	1.2857	0.4000
31-1-2	4.85	32	40	13	1.2500	0.4063
32-7-2	0.85	32	41	15	1.2813	0.4688

J_{sr}, isothermal remanent saturation magnetization per volume; H_{cr}, remanent coercivity force; H'_{cr}, acquisition remanent coercivity force; $H_{1/2I}$, median destructive field.

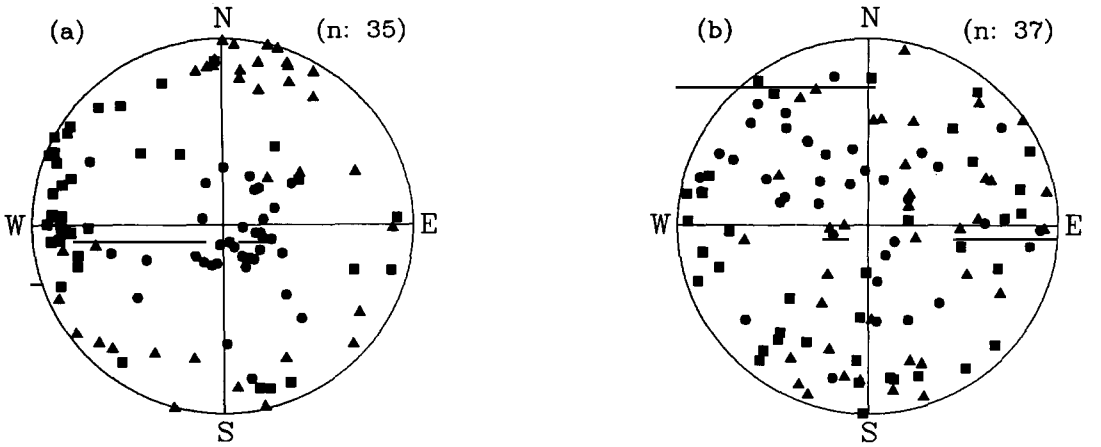


Fig. 9. Equal-area, lower hemisphere projections of specimen AMS axes for (a) the Hekou Formation and (b) the Jieshan Formation (after tilt correction). ■, the maximum principal susceptibility axes K_{max}; ▲, the intermediate principal susceptibility axes K_{int}; ●, the minimum principal susceptibility axes K_{min}. n represents the number of specimens used.

anisotropy P ($K_{max}/K_{min} < 1.05$; e.g. Hrouda, 1982) and minimum susceptibility axes perpendicular to the bedding. If the rocks suffered post-depositional alteration, however, this primary sedimentary fabric may be disturbed, overprinted or even obliterated by secondary fabrics. As alterations in rock units (such as compaction, penetrative deformation or metamorphism) can affect pre-existing remanent magnetization (RM) in one way or another (such as inclination shallowing, alteration of remanence intensity, as well as/or direction and overprinting), studies of magnetic anisotropy may help in detecting these alterations and therefore provide additional clues to the origin of remanence. An example of such a study is that of Tauxe et al. (1990), for the Siwalik redbeds in northern Pakistan. These workers found that sediments which preserved a primary remanence tend to have a more primary fabric, whereas those that retained little or no primary component or acquired a coherent post-depositional component of remanence tend to have a more altered fabric. This phenomenon is also found in the present low-field AMS investigation. Figures 9 and 10 and Table 3 illustrate the AMS results from specimens that carry characteristic components in both the Hekou and the Jieshan Formations.

5.1. Hekou Formation

The degree of anisotropy P from the Hekou Formation is very low, mostly less than 1.05 (Table 3), as expected for a sedimentary fabric. In addition to low anisotropy, the minimum susceptibility axes K_{min} are approximately perpendicular to the bedding (Fig. 9(a)) and, consequently, the specimen imbrication angle (the angle between the magnetic foliation and bedding) is generally close to or less than 20° (Fig. 10), in agree-

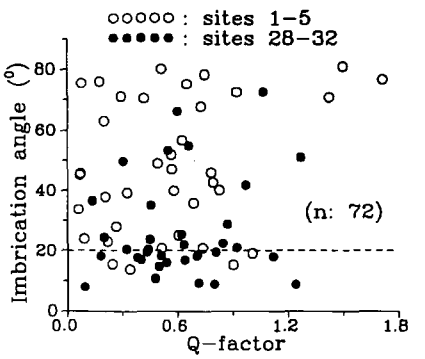


Fig. 10. Plot of Q-factor vs. imbrication angle A, where Q-factor equals $(K_{max}\text{-}K_{int})/[\frac{1}{2}(K_{max} + K_{int})\text{-}K_{min})]$ (Tarling, 1983) and A is the angle between the K_{min} axis and the bedding pole.

ment with the behaviour of primary sedimentary fabrics (e.g. Rees, 1966; Hamilton and Rees, 1970; Ellwood, 1979; Hrouda, 1982).

The above AMS investigation indicates that specimens from the Hekou Formation have a primary sedimentary fabric. Absence or sparse occurrence of hematite/pyrrhotite (as indicated by the IRM measurements) as well as lack of tectonic/metamorphic textures and abundance of depositional laminations (as shown by field and microscopic observations) suggest little or no post-depositional alteration for the Hekou Formation. Accordingly, a primary remanence, if it is carried by magnetic grains with sufficient relaxation time to resist thermal fluctuations, is expected to remain in these rocks.

Compaction of sediments can contribute to inclination shallowing up to the order of tens of degrees. This has been demonstrated in laboratory studies (e.g. Blow and Hamilton, 1978; Anson and Kodama, 1987; Deamer and Kodama, 1990; Lei and Banerjee, 1990), in deep-sea sediments (e.g. Kent and Spariosu, 1982; Celaya and Clement, 1988; Arason and Levi, 1990a) or in numerical modelling (e.g. Arason and Levi, 1990b). However, this compaction-induced inclination shallowing appears to be insignificant in the Hekou Formation. Arguments in favour of this interpretation include the following:

(1) the E-factor of each site in the Hekou Formation is only slightly larger than unity (mostly less than 1.01; see Table 3), indicating that the susceptibility ellipsoid is only slightly oblate. This suggests little influence of compaction on the depositional alignment of magnetic minerals, otherwise a more oblate ellipsoid would be expected;

(2) In a recent laboratory experiment of redeposition of natural sediments, Lei and Banerjee (1990) proposed that a key factor affecting inclination shallowing is the net-magnetic-moment over mass (M/m) ratio. In large low-magnetic-moment multidomain magnetite particles, inclination shallowing might be produced by dominating gravitational torques causing particle rotations about horizontal axes. In contrast, in the case of slow deposition of isolated submicron magnetite grains, magnetic alignment torques will dominate over gravitational forces, and no inclination shallowing should be expected. As discussed above, the characteristic component is carried by SD/PSD-type titanomagnetite grains which have a high M/m ratio; therefore, one can expect an insignificant inclination shallowing, or none at all, in the Hekou Formation.

Based on the primary sedimentary fabric preserved in the rocks, the predominant contribution of SD/PSD titanomagnetite to the remanence, the normal polarity consistent with the Cretaceous normal superchron, the pre-folding age of magnetization and negligible inclination shallowing, we believe that the characteristic remanence represents a faithful record of the geomagnetic

TABLE 3

Site-mean parameters of magnetic anisotropy and associated standard deviations

Site	N	$K_m \pm \Delta K_m$	$P \pm \Delta P$	$L \pm \Delta L$	$F \pm \Delta F$	E-factor
1	7	0.9868 ± 0.0255	1.0989 ± 0.0441	1.0455 ± 0.0258	1.0511 ± 0.0320	1.0054
2	8	0.8874 ± 0.0519	1.0933 ± 0.0567	1.0387 ± 0.0234	1.0520 ± 0.0334	1.0128
3	8	1.2351 ± 0.0843	1.0843 ± 0.0447	1.0332 ± 0.0187	1.0501 ± 0.0534	1.0164
4	7	0.6583 ± 0.0809	1.2694 ± 0.0992	1.0576 ± 0.0378	1.2009 ± 0.0932	1.1355
5	7	0.6132 ± 0.2298	1.1898 ± 0.1166	1.0423 ± 0.0179	1.1425 ± 0.1211	1.0961
28	6	3.7205 ± 1.7569	1.0636 ± 0.0458	1.0228 ± 0.0118	1.0398 ± 0.0423	1.0166
29	6	0.7086 ± 0.0775	1.0697 ± 0.0403	1.0287 ± 0.0166	1.0396 ± 0.0247	1.0106
30	8	27.3610 ± 6.1680	1.0343 ± 0.0029	1.0167 ± 0.0022	1.0172 ± 0.0041	1.0003
31	7	6.1321 ± 0.8077	1.0312 ± 0.0091	1.0121 ± 0.0029	1.0189 ± 0.0100	1.0067
32	8	1.6670 ± 0.0797	1.0356 ± 0.0100	1.0139 ± 0.0085	1.0215 ± 0.0085	1.0075

N, number of specimens measured per site; K_m and ΔK_m, mean susceptibility and standard deviation (unit = 1×10^{-4} SI (vol)$^{-1}$); P and ΔP, degree of anisotropy (K_{max}/K_{min}) and standard deviation; L and ΔL, magnetic lineation (K_{max}/K_{int}) and standard deviation; F and ΔF, magnetic foliation (K_{int}/K_{min}) and standard deviation; E-factor: $K_{int}^2/(K_{max} \cdot K_{min})$.

field when the sediments were deposited and that it is therefore a depositional remanent magnetization (DRM) in origin.

5.2. *Jieshan Formation*

The AMS of this formation is significantly different from that of the Hekou Formation. Major differences include the following: (1) the degree of anisotropy P is much higher (mostly larger than 1.1); (2) the shape of the susceptibility ellipsoid is much more oblate for most specimens, as indicated by higher values of the E-factor; (3) the specimens have randomly directed rather than subvertical K_{min} axes (Fig. 9(b)) and, as a result, the specimen imbrication angles are mostly located in the 40–80° range (Fig. 10). These significant differences clearly indicate destruction of a depositional fabric and development of a more altered fabric in the Jieshan Formation, otherwise the differences would be minor because the magnetic mineralogy is similar to that of the Hekou Formation. Consequently, the magnetic fabric of this formation can be regarded unmistakeably as a secondary fabric.

A secondary fabric may result from deformation or from metamorphism, or from both. Absence of metamorphism in the rock unit (Fujian Bureau of Geology, 1985) suggests a deformational origin of the observed fabric, which is consistent with the deformation textures (for example, slickensides and fractures) observed on the outcrops. The first phase of tectonic activity seems to be mainly responsible for the deformation of this rock unit, as it is the most important tectonic event during Cretaceous time.

This highly disturbed fabric suggests that the depositional alignment of magnetic minerals has been altered and, as a consequence, a 'primary' remanence is unlikely to be preserved in the rock unit. This can also be inferred from the following discussion. Assuming that the characteristic remanence of the Jieshan Formation represents a 'primary' remanence, the sampling area would be located at 2.4°S during the early stage of the Early Cretaceous. According to the primary remanence that we obtained from the Hekou Formation, the sampling area was situated at 18.3°N by the late stage of the Early Cretaceous. This means that the sampling area experienced a 21° northward displacement during the early stage of the Early Cretaceous and the late stage of the Early Cretaceous. Such a high rate of latitudinal movement, if it occurs, can only be related to the subduction of an ocean between the Huanan and Yangzi blocks. However, there is no geological evidence to support the existence of a wide Early Cretaceous ocean between them. Therefore, the characteristic remanence of the Jieshan Formation represents a secondary magnetization instead of a primary one.

The secondary magnetization is more probably related to a viscous thermal remanent magnetization (VPTRM) than to a chemical remanent magnetization (CRM). Arguments in favour of this include the following: (1) Hematite and pyrrhotite are often found in hydrothermal processes which are responsible for CRM. However, these minerals are absent or only sparsely present in the samples, as indicated by the IRM measurements and microscopic observations. (2) Since Cretaceous time, volcanic and magmatic activities have been weak in western Fujian; however, they are still very intense and extensive in eastern Fujian (Fujian Bureau of Geology, 1985). On the one hand, this difference in activities greatly reduces the possibility of occurrence of CRM in western Fujian, as a result of the lack of hydrothermalism, and, on the other hand, the large-scale igneous activities in eastern Fujian may elevate the geothermal gradient in western Fujian to a higher level, which in turn increases the possibility of acquiring a partial thermal remanent magnetization (PTRM). (3) During the first phase of tectonic activity between the Early and Late Cretaceous, western Fujian subsided to form a series of subsidence basins where Late Cretaceous redbeds, with a total thickness of over 2000 m, were deposited. This subsidence, as well as the igneous activities in eastern Fujian, might have allowed the Jieshan Formation to be subjected to a moderate burial temperature over a prolonged period and thus to acquire a moderate temperature viscous remanence. This burial remagnetization may have remained fossilized in the rocks during the large-scale uplift of the region at the end of the Cretaceous.

It is worth mentioning that a similar secondary direction has also been found in a continuing paleomagnetic study of the Early Triassic in the same area ($D = 16.4°$, $I = 42.2°$, $\lambda_p = 75.2°$N, $\phi_p = 209.3°$E, A_{95} (95% confidence circle of site mean or unit mean pole) = 4.6°, in geographic coordinates; the result of this study will be published elsewhere). This evidence indicates that the region investigated might have suffered a systematic widespread remagnetization during the late stage of the Yanshanian orogenic activities. Remagnetization events associated with the Yanshanian orogeny are also documented elsewhere in eastern China. For example, in a paleomagnetic survey of Phanerozoic rocks of seven provinces in the South China and Sino-Korean Blocks, Zhao and Coe (1984, 1985) and Zhao (1987) found a prevalence of Late Mesozoic overprinting related to the Yanshanian orogeny in the Late Mesozoic; Kent et al. (1987) found that the marine platform carbonates and clastics, ranging in age from Early Cambrian to Early Triassic at Nanjing (Jiangsu Province) and Yichang (Hubei Province) are typically overprinted by Late Meso-

TABLE 4

Cretaceous poles from Huanan and other blocks of eastern China

Tectonic unit	No.	Age	λ_p (°N)	ϕ_p (°E)	$\kappa(K)$	α_{95} (A_{95})	Test	References
Huanan Block								
Shanghang, Fujian	H1	K_2	67.9	186.2		9.2	F	Hu (1986)
Ninghua, Fujian	H2	K_2	84.1	170.2		6.5		Hu (1986)
Shaxian, Fujian	H3	K_2	72.8	184.9		4.2		Hu (1986)
Hong Kong	H4	K_2	78.2	171.9	26.1	8.7	R	Chan (1991)
Mean	M_1	K_2	75.8	181.2	(125.9)	(8.2)		H1–H4
Yongan, Fujian	H5	K_1	66.9	221.4	(203.3)	(5.4)	F	This study
Gyeongsang, S. Korea	H6	K_1	67.6	205.1	34.6	(5.8)	FR	Lee et al. (1987)
Yangzi Block								
Nanjing, Jiangsu	Y1	K_2	76.3	172.6	(23.0)	(10.3)	F	Kent et al. (1986)
Dianzhong, Yunnan	Y2	K_2	69.0	188.0		7.5		Liang et al. (1987)
Ya'an, Sichuan	Y3	K_2	83.0	188.7	54.6	8.4		Zhuang et al. (1988)
Ningbo, Zhejiang	Y4	K_2	71.4	185.4	193.0	8.9		Feng et al. (1989)
Panxi, Sichuan	Y5	K_2	78.9	186.6	(322.0)	(6.9)		Zhu et al. (1988)
Mean	M_2	K_2	75.8	184.2	(191.4)	(5.5)		Y1–Y5
Ya'an, Sichuan	Y6	K_1	80.4	208.2	45.8	7.3		Zhuang et al. (1988)
Ya'an, Sichuan	Y7	K_1	78.6	201.0	47.9	15.3		Zhuang et al. (1988)
Dianzhong, Yunnan	Y8	K_1	71.0	201.3		7.8		Liang et al. (1987)
Zhejiang	Y9	K_1	76.2	225.7	(662.0)	(4.8)	R	Lin and Fuller (1985)
Ya'an, Sichuan	Y10	K	70.9	225.2	63.1	3.6	FR	Enkin et al. (1991)
Mean	M_3	K_1	75.7	213.0	(226.0)	(5.1)		Y6–Y10
Sino-Korean Block								
Datong, Shanxi	S1	K_2	79.2	170.7		5.8	R	Zheng et al. (1991)
North Korea	S2	K_2	69.0	182.0	193.0	4.0		Gurarii et al. (1966)
Zhucheng, Shandong	S3	K_1	69.0	200.9	40.0	12.2	R	Lin and Fuller (1985)
Hainan Island								
Dongnanjiang	I1	K_2	80.6	175.1		5.2		Mo and Shi (1987)
Eastern China								
	M_4	K_2	76.0	181.5	(188.3)	(3.2)		H1–H4, Y1–Y5, S1–S2, I1
	M_5	K_1	72.8	211.3	(174.8)	(4.2)		H5, H6, Y6–Y10, S3

λ_p (ϕ_p), latitudes (longitudes) of paleomagnetic poles; κ (K), precision parameters of mean directions (poles); α_{95} (A_{95}), 95% confidence circles of mean directions (poles); R, reversal test; F, fold test; K_1, Early Cretaceous, K_2, Late Cretaceous.

zoic to Recent fields, and thus attributed the overprinting to the Yanshanian orogenic activities in South China. The secondary magnetization in the Jieshan Formation further confirms the occurrence of widespread Late Mesozoic remagnetization events in South China.

6. Discussion and conclusion

6.1. Cretaceous poles from Huanan and other blocks of eastern China

Given the assumption of a geocentric dipole field, the tectonic relationships between two blocks can be inferred by comparison of their corresponding pole positions or APWPs. To aid the determination of the tectonic relationship between the Huanan and Yangzi Blocks, the assemblage of eastern China (referring to an area that includes the Sino-Korean, Yangzi and Huanan Blocks as well as Hainan Island) and the relative motion of eastern China with respect to Eurasia during the Cretaceous time, we have collected all available Cretaceous poles from these blocks and evaluated them in detail. The poles are listed in Table 4 only if they (1) are well-determined paleomagnetically, (2) are known to have an age of the Early or Late Cretaceous, (3) have been proved by field tests (such as the fold test and the reversal test) to be primary, or are consistent with coeval poles from other areas of the same block with positive field tests, and (4) are determined from areas where structural controls are unambiguous.

6.1.1. Huanan Block

Six Cretaceous poles are available from the Huanan Block (H1–H6), including the result obtained in the present study. H1–H3 were reported by Hu (1986) from Late Cretaceous redbeds in western Fujian. The magnetization was believed to be primary or to be acquired shortly after deposition during diagenesis. Arguments in favour of this age of acquisition are the following: (1) the magnetization shows normal polarity corresponding to the Cretaceous normal superchron; (2) H1 passes the fold test at 95% confidence level (the rock units were folded at the end of the Cretaceous), and (3) the three poles are consistent, although they were obtained from different localities separated by over 80 km. Chan (1991) reported a pole from two Late Mesozoic granitic plutons in Hong Kong (H4). This pole is claimed to be of Late Cretaceous age because of the presence of a dual polarity and cooling ages of the granites. Hsu (1987) obtained a pole ($\lambda_p = 66.0°$, $\phi_p = 221.5°$, $\alpha_{95} = 3.9°$) from a Late Cretaceous redbed in Guangdong Province. This result has a dual polarity. However, it is paleomagnetically distinguishable from other Late Cretaceous poles for the Huanan Block or other blocks in eastern China but indistinguishable from Early Cretaceous poles. This difference may be attributed either to a greater age (Early Cretaceous?) of the rock unit studied or to a clockwise rotation of about 10° of the sampling area with respect to the rest of Huanan Block. As insufficient information is available to evaluate its reliability, we tentatively excluded this pole from our list unless other evidence becomes available to prove its reliability. H6 was reported from volcanic and sedimentary rocks of Early Cretaceous age in the Gyeongsang Basin, south of the Ogcheon Deformed Belt in South Korea (Lee et al., 1987). The presence of a dual polarity and a positive fold test indicate that the primary remanence has been isolated. According to Hsü's model, the area south of the Ogcheon Deformed Belt in South Korea may be the northeastern extension of the Huanan Block (Hsü et al., 1990); therefore, this result is included in the Huanan category. Earlier, Otofuji et al. (1983) reported a pole ($\lambda_p = 69°$, $\phi_p = 191°$, $\alpha_{95} = 8.3°$) from Cretaceous rocks in the same basin of South Korea. Lee et al. (1987) pointed out that this result is an overprinting, as a result of insufficient demagnetization. Therefore, this pole is not included in the list. Li et al. (1982) obtained a result from Early Cretaceous redbeds in a two-ring-shaped structure in southern Jiangxi (inner ring: $\lambda_p = 76.0°$, $\phi_p = 170.2°$, $\alpha_{95} = 2.8°$; outer ring: $\lambda_p = 68.2°$, $\phi_p = 186.6°$, $\alpha_{95} = 2.6°$). The poles reported in this study are closer to Late Cretaceous poles for this block than to Early Cretaceous ones. Unfortunately, no field tests are available to eval-

uate their reliability; however, at least two possibilities exist: (1) the sampling area has been rotated anticlockwise about 10° with respect to the rest of the Huanan Block. This is not impossible because the sampling sites are located in a locally deformed structure; (2) the rocks sampled are older than those in our study and that of Lee et al. (1987). One argument in favour of the second interpretation is that some sites (e.g. in Chongshiyu) have been suspected by the authors to be of Late Cretaceous age. In view of the uncertainties, we do not use this result for further analysis.

6.1.2. Yangzi Block

Cretaceous poles are well determined for this block. Y1 was reported by Kent et al. (1986) from redbeds at Nanjing, in the eastern Yangzi Block. This result passed the fold test at 95% confidence level. In the same study, Kent et al. reported another Late Cretaceous pole from redbeds at Huidong, Sichuan, in the western Yangzi Block. This pole ($\lambda_p = 80.8°$N, $\phi_p = 296.8°$E, $A_{95} = 7.7°$) is statistically different from that obtained for Nanjing. As a local rotation might have been involved in southwest Sichuan, as suggested by Kent et al. (1986) and by Enkin et al. (1991), we prefer to exclude this pole from our list until a reliable tectonic correction can be made on it. No field tests are available on the remaining four Late Cretaceous results (Y2–Y5). However, in view of the fact that the PEF overprinting has been successfully removed, as described by Kent et al. (1986) and that these results are highly consistent with Y1, we believe that they can be trusted. Four Early Cretaceous poles (Y6–Y9) are available for the Yangzi Block. Although no field tests are available on these results, the careful demagnetization procedures described by Kent et al. (1986) and Enkin et al. (1991) the wide distances between the localities of these studies and a high degree of consistency of the poles suggest that the primary remanence has been isolated. More recently, Enkin et al. (1991) reported a tectonically corrected Cretaceous pole (Y10) from continental redbeds ranging in age from Upper Jurassic to Eocene at Ya'an, in the western Sichuan Province. A positive fold test and a rock magnetic study indicate that the remanence is primary. This pole was obtained from an area where Enkin et al. (1991) believed that a 15° counter-clockwise (CCW) rotation has taken place. It is interesting to note that before correcting for this rotation, the pole position ($\lambda_p = 78.6°$, $\phi_p = 273.4°$, $\alpha_{95} = 3.6°$) is indistinguishable from that obtained by Kent et al. (1986) for Huidong, Sichuan; therefore, the latter also seems to have suffered a 15° CCW rotation, as suggested by Enkin et al. (1991). The time of acquisition of remanence is thought to be within 150–50 Ma; however, the exact age might be biased towards the Early Cretaceous because the tectonically corrected pole position is much closer to Early Cretaceous poles than to Late Cretaceous poles for this block (see Table 4) and because 19 of 26 sites used in the study have an age of Late Jurassic to Early Cretaceous (160–110 Ma; see Fig. 11 of Enkin et al. (1991)). In the light of this observation, we tentatively regard this pole as an Early Cretaceous pole.

6.1.3. Sino-Korean Block

Very few Cretaceous poles (S1–S3) are available for the Sino-Korean Block. S1 was reported by Zheng et al. (1991) from Late Cretaceous limestones at Datong (Shanxi Province). This result shows a dual polarity. Gurarii et al. (1966) obtained a Late Cretaceous pole from sandstones in North Korea. Unfortunately, no field test is available. Although indistinguishable from S1, we prefer to use the latter as the representative pole because of the presence of dual polarity and the use of modern demagnetization procedures. Lin and Fuller (1985) obtained an Early Cretaceous pole (S3) from andesites in Shandong Province. This result has a dual polarity and is regarded as primary. Another Early Cretaceous pole was reported recently by Cheng et al. (1988) based on two sandstone sites at Yan'an (Shanxi Province). This result ($D = 357.1°$, $I = 50.9°$, $\alpha_{95} = 4.2°$, $\lambda_p = 84.6°$, $\phi_p = 310.2°$) is statistically distinguishable from S1–S3 and from other Cretaceous poles elsewhere in China. As the site mean direction before tilt correction (S16: $D = 6.3°$, $I = 53.5°$; S17: $D = 13.9°$, $I = 56.3°$) is close to the PEF of the locality ($D = 357°$, $I = 54°$) and no field tests were performed to constrain the age of magneti-

zation, the possibility of PEF overprinting cannot be ruled out, and therefore we do not use this result for further analysis.

6.1.4. Hainan Island

According to the traditional view of geology of South China, Hainan Island is regarded as a Hercynian fold belt within the South China Fold System (Huang, 1984; Yang et al., 1986). In Hsü's model, however, it is regarded as the northern promontory of the Dongnanya Block, which was separated from the Huanan Block by the Gunanhai Ocean during the Late Paleozoic and Early Mesozoic. Paleomagnetic data for this island are very limited, and only one Late Cretaceous result (I1) is available. This result was reported by Mo and Shi (1987) from redbeds at Dongnanjiang. This pole is indistinguishable from other Cretaceous poles for eastern China. Although no field tests are available to evaluate its reliability, we intend to retain this pole in our list at this stage.

When sufficient data are available, mean poles for each block (M_1, M_2 and M_3, Table 4) can be calculated for a geological epoch and regarded as representative of this epoch.

6.2. Tectonic relationships between the Huanan and Yangzi Blocks during Cretaceous time

In Fig. 11, we compare our result and other Cretaceous poles for the Huanan Block with the Yangzi APWP. The latter is a refined APWP of Lin and Fuller (1990). As their Cretaceous poles were based on only three studies, we prefer to replace them with the recalculated mean poles M_2 and M_3 (Table 4) in the APWP because the two mean poles are based on 10 consistent results from various parts of the Yangzi Block and are therefore more representative.

The most striking observation is that our result and other Huanan Cretaceous studies are in agreement with the Cretaceous APWP segment,

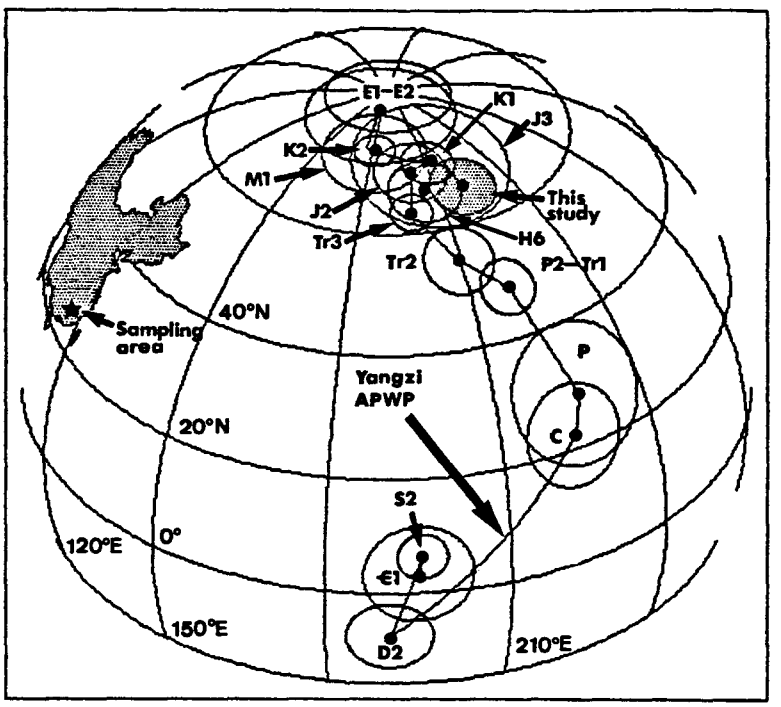


Fig. 11. Comparison of the pole from the present study (marked with a shaded circle) and other Cretaceous poles (M_1 and H6 in Table 4) with the Yangzi APWP (orthographic azimuthal projection). Star represents the sampling area. E_{1-2}, Paleocene–Eocene; K_2, Late Cretaceous; K_1, Early Cretaceous; J_3, Late Jurassic; J_2, Middle Jurassic; J_1, Early Jurassic; Tr_3, Late Triassic; Tr_2, Middle Triassic; P_2–Tr_1, Permo-Triassic; P, Permian; C, Carboniferous; D_2: Middle Devonian; S_2, Middle Silurian; ϵ_1, Early Cambrian.

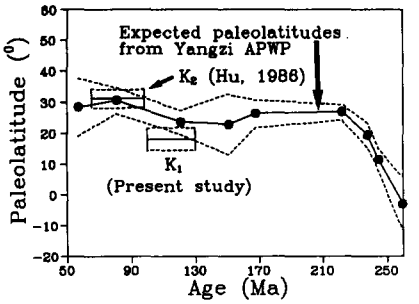


Fig. 12. Comparison of the observed and expected paleolatitudes of the sampling area. Solid curve represents paleolatitudinal changes of the sampling area expected from the Yangzi APWP; dashed curves represent its 95% confidence limits. The observed paleolatitudes of the sampling area during Early and Late Cretaceous times are given by small rectangles; the vertical lines represent associated 95% confidence intervals and the horizontal ones represent the age of magnetization.

as indicated by an overlap of their 95% confidence circles. This implies that the Huanan Block has been attached to the Yangzi Block at least since the Early Cretaceous, in agreement with Hsü's model, which suggests a Middle Triassic age for the initial collision of the two blocks (Hsü et al., 1990). On close inspection, however, we found that our pole corresponds to a slightly lower paleolatitude (18.3° ± 3.6°) than that expected (23.7° ± 3.9°) from the Yangzi APWP (Fig. 12). As 95% confidence limits of the observed and expected paleolatitudes only partly overlap, this latitudinal discrepancy may indicate a real feature. As pointed out above, the compaction-induced inclination shallowing is negligible; therefore, this discrepancy may be indicative of a small northward movement of the sampling area with respect to the Yangzi Block. This interpretation agrees with Hsü's model, and suggests that after the collision of the Huanan Block with the Yangzi Block, the Banxi Mélange and Huanan Flysch Nappes were overthrust hundreds of kilometres northwards onto the Yangzi carbonate platform until they were tectonically superimposed on Jurassic and Lower Cretaceous rocks (Hsü et al., 1988a, b, 1990; Sengör et al., 1988; Li et al., 1989). Some geological evidence has been found to support this Mesozoic overthrust tectonics in South China. For example, a recent deep-sounding seismic profile across the Xuefeng Mountains has demonstrated a nearly flat reflecting horizon several kilometres below the surface, which, according to Hsü et al. (1990), represents a flat thrust between the Banxi Mélange–Huanan Flysch Nappes and the Yangzi Deformed Belt. In the Yangzi Deformed Belt, several tectonic windows have been identified at Lantian in southern Anhui, West Hill to the east of Nanchang and Leping to the west of Nanchang. This clearly shows the thrust contact between the Banxi Mélange and Yangzi Paleozoic carbonates (Hsü et al., 1988a, b, 1990; Li et al., 1989). In addition to the above-mentioned flat thrust belt and tectonic windows, Precambrian granites capping high peaks of several mountains in the Yangzi region, such as Yuanbaoshan in Guangxi, Fanjingshan in Guizhou, Xuefeng in Hunan and Jiulingshan in Jiangxi, have been interpreted as klippes and might have been displaced hundreds of kilometres from their original location in the Huanan Block (Hsü et al., 1988a, b, 1990; Li et al., 1989). From this geological evidence, one may therefore expect a relatively lower depositional paleolatitude for the Hekou Formation than that predicted from the coeval pole position of the Yangzi Block. As the Late Cretaceous poles obtained by Hu (1986) from other areas in western Fujian correspond to the same paleolatitude as that predicted by the coeval pole of the Yangzi Block (Fig. 12), the overthrust of the Huanan Block above the Yangzi Block might have been completed by the Late Cretaceous. This is again consistent with the geological observation that the youngest sediments found beneath the overthrusts are Early Cretaceous redbeds (Hsü et al., 1990).

To summarize, the Huanan Block has been accreted to the Yangzi Block at least since the Early Cretaceous. A slight discrepancy between the observed paleolatitude and that expected from the Yangzi APWP might be indicative of a small northward displacement of the sampling area by an advancing rigid Huanan basement nappe during the Early–Late Cretaceous. The exact amount of the displacement, however, cannot be determined paleomagnetically at this stage because it

falls within the limit of error of the measurement. Additional Mesozoic poles from the Huanan Block are necessary to confirm our interpretation.

6.3. *Assemblage of eastern China*

The very limited paleomagnetic data from Hainan Island restrict a detailed discussion on its tectonic relationship with the Huanan Block. However, as can be seen from Fig. 13, the Late Cretaceous pole of Hainan Island is coherent with those for the Huanan, Yangzi and Sino-Korean Blocks, suggesting that the Dongnanya Block, or at least its northern promontory (i.e. Hainan Island), has been fully sutured to the Huanan Block since the Late Cretaceous. This coincides with the proposal that the former was attached to the latter before the Cretaceous (Hsü et al., 1990). A close inspection of Fig. 13 indicates that all the Cretaceous poles are in agreement; therefore, we suggest that the whole of eastern China was fully assembled by Cretaceous time, although a small northward overthrust of the Huanan Block above the Yangzi Block may have occurred between the Early and Late Cretaceous. This paleomagnetic observation may correlate to a plate reorganization of the West Pacific during the Cretaceous, as suggested by Hsü et al. (1990).

6.4. *Post-Cretaceous movement of eastern China with respect to Eurasia*

This problem has been discussed by many paleomagnetists, who have put forward various proposals. The points are summarized as follows.

(1) North and South China were displaced thousands of kilometres eastward with respect to Eurasia, as a result of the tectonic extrusion of Asia during the India–Eurasia convergence in Cenozoic time (Lin and Fuller, 1985; Lin, 1987).

(2) A sinistral displacement of over 1000 km could have occurred on the NNE–SSW trending

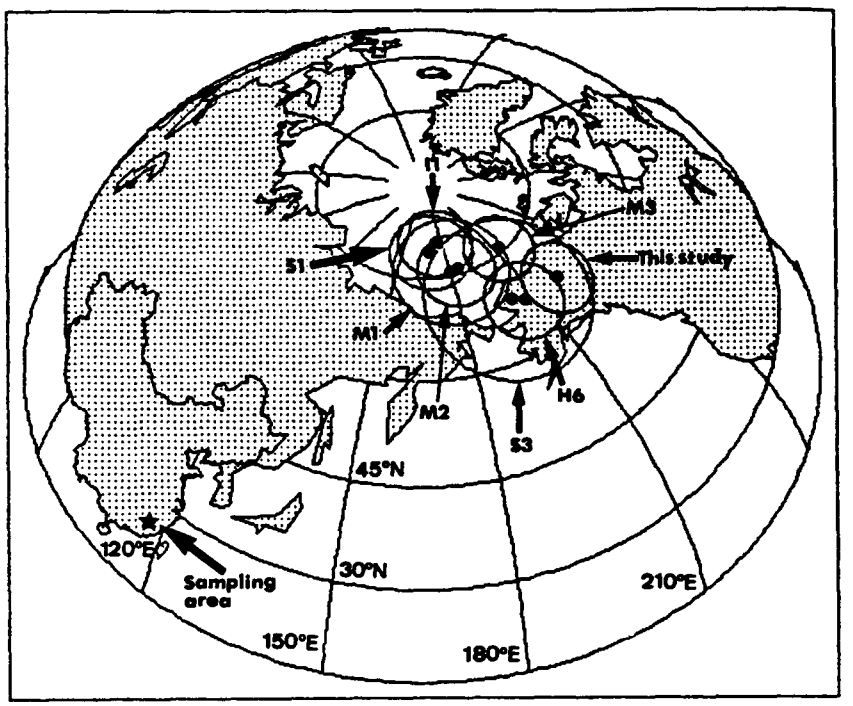


Fig. 13. Cretaceous poles from various blocks of eastern China. The sampling area is shown by a star. (For abbreviations, see Table 4 (orthographic azimuthal projection).)

Tancheng–Lujiang Fault since the Late Cretaceous (Courtillot and Besse, 1986; Lee et al., 1987).

(3) North and/or South China was/were fully sutured to Eurasia by the end of the Late Jurassic (Zhao et al., 1990; Enkin et al., 1991) or Cretaceous (Achache et al., 1983; Kent et al., 1986; Lee et al., 1987; Zhai and Zhou, 1989; Chan, 1991; Zheng et al., 1991) without significant subsequent relative motion.

Two reasons are responsible for the divergence of the above-mentioned proposals. First, different Eurasian reference APWPs were used for paleomagnetic comparison. These include the APWPs from Irving and Irving (1982) (Lin and Fuller, 1985), from Irving (1977) (Courtillot and Besse, 1986), from Westphal et al. (1986) (Zhai and Zhou, 1989) or from Sasajima and Maenaka (1989) (Zheng et al., 1991). Second, different Euler poles were chosen for tectonic analysis; for example, an Euler pole (68.5°N, 31.5°E) close to the North Pole was used by Lin and Fuller (1985), who inferred eastward motion of eastern China, whereas another Euler pole (16°N, 7°E) was chosen by Courtillot and Besse (1986) to suggest left-lateral displacement on the Tancheng–Lujiang Fault. Between these two reasons, the first may be more important, because any reasonable interpretation of relative motions between China and Eurasia relies not only on the reliability of Chinese data but also on that of the Eurasian APWP. As to the reliability of Chinese data, we believe that the results listed in Table 4 can be trusted because nine of the 20 poles passed the fold or reversal test and all of them display a high consistency. The reliability of Eurasian APWPs has been discussed by several workers (Westphal et al., 1986; Lee et al., 1987;

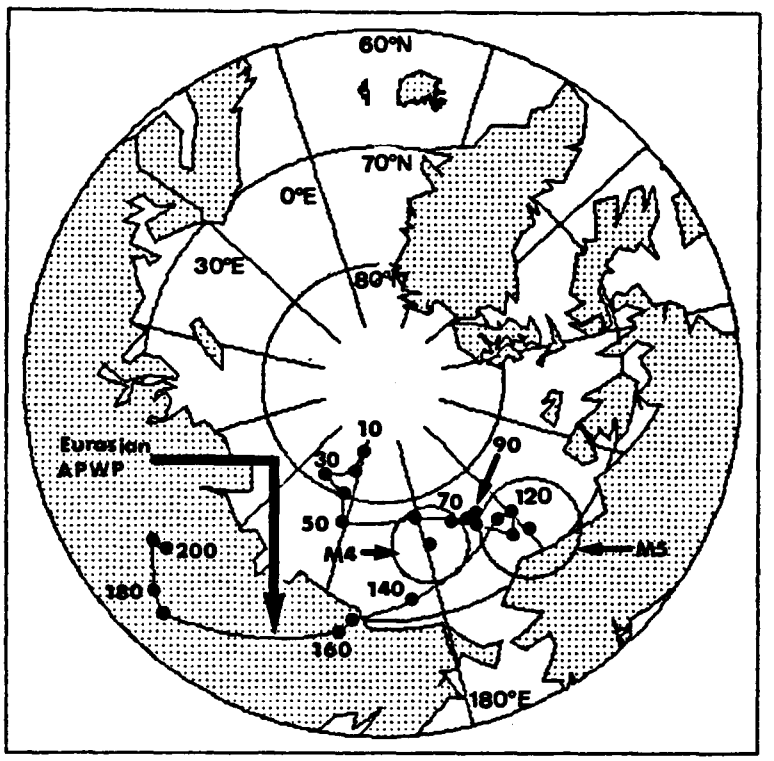


Fig. 14. Comparison of Cretaceous poles (M_4: Late Cretaceous pole; M_5: Early Cretaceous pole) of eastern China with the Eurasian reference APWP of Besse and Courtillot (1991). Numbers next to the small solid circles represent ages (in Ma) of the poles (Lambert equal-area conical projection).

Besse and Courtillot, 1988, 1991). They pointed out that Soviet data may be in error, and that previous APWPs (Irving, 1977; Irving and Irving, 1982) should be revised because they involved little selection of Soviet data and inclusion of data from orogenic regions. They also suggested that Southeast Asian poles (at least since the Early Cretaceous) should be included when constructing an APWP for stable Eurasia. Based on these considerations, Besse and Courtillot (1991) recently published a revised and synthetic APWP for Eurasia which is probably the most reliable APWP for stable Eurasia to date. Therefore, we use this path as a reference for discussion of the relative motion between eastern China and Eurasia.

As discussed above, eastern China may be regarded as a single tectonic unit since Cretaceous time. Therefore, for this period, the poles from the Huanan, Yangzi and Sino-Korean Blocks, as well as Hainan Island, can be taken together to determine global means for the whole of eastern China. The poles thus obtained for the Early and Late Cretaceous are given in Table 4. Both of them have a very high precision and thus represent the best-defined Cretaceous poles for eastern China to date. As can be seen in Fig. 14, these two Cretaceous poles correspond to a Cretaceous hairpin turn (from approximately 140° to 60°) in the revised Eurasian APWP of Besse and Courtillot (1991). Enkin et al. (1991) also reported this phenomenon in a study of Upper Jurassic–Lower Tertiary redbeds from Sichuan, Yangzi Block, and concluded that the Mesozoic hairpin turn in Besse and Courtillot's (1991) APWP is a real feature. At present, all Cretaceous poles from eastern China confirm this conclusion. The coincidence between the Cretaceous poles for eastern China and the Cretaceous hairpin turn in the Eurasian APWP implies that the whole of eastern China was fully sutured to Eurasia by this period. This is clearly demonstrated in Fig. 15, which shows that no paleomagnetically resolvable rotation and latitudinal displacement have taken place between eastern China and Eurasia since Cretaceous time. Previous geological and tectonic studies (Molnar and Tapponnier, 1975; Tapponnier and Molnar, 1977; Tapponnier et al., 1982, 1986) proposed an eastward lateral extrusion to accommodate the post-collisional India–Eurasia convergence. According to the suggestion of those workers, southern China has been displaced substantially eastward with respect to Eurasia since the indentation of India into Eurasia. This is in contradiction with the above-mentioned paleomagnetic results; also, this eastward lateral extrusion is believed to be unlikely according to several recent studies. For example, Kidd and Molnar (1988) pointed out that the left-lateral motion of China on faults such as the Altyn Tagh, Kunlun and Kang Ting Faults, which have been argued to be the main

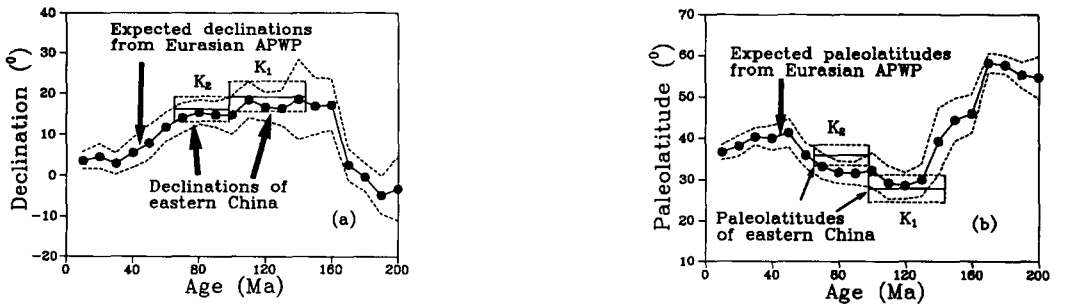


Fig. 15. Comparison of the observed as well as expected declinations (a) and paleolatitudes (b) at a central reference point (32°N, 112°E) of eastern China. Solid curve represents the declinational (paleolatitudinal) changes of the central reference point expected from the Eurasian APWP; dashed curves are the associated 95% confidence limits. The observed declination or paleolatitude of the central reference point during Early and Late Cretaceous times are given by small rectangles; the vertical lines represent their associated 95% confidence intervals and the horizontal ones represent the age of magnetization.

loci of escape, is relatively small, and far less than the thousands of kilometres required by the extrusion hypothesis. Dewey et al. (1989) also concluded that lateral extrusion was not a major factor in accommodating the India–Eurasia convergence, and suggested that a maximum eastward extrusion of the convergent zone was only about 250 km. More recently, England and Molnar (1990) concluded that the image of 'lateral extrusion' caused by the India–Eurasia collision is an illusion, and pointed out that little material is moving eastward out of India's path into Eurasia by left-lateral simple shear. Clearly, these recent studies are consistent with the present paleomagnetic conclusion. One may therefore conclude that the indentation of India into Eurasia did not cause any paleomagnetically resolvable movement of eastern China with respect to Eurasia.

Obviously, if one accepts the Eurasian reference APWP of Besse and Courtillot (1991), the first proposal indicating that eastern China was displaced thousands of kilometres eastward since the India–Eurasia collision lacks a paleomagnetic basis, considering the absence of discrepancy between the Cretaceous poles of eastern China and those of Eurasia. For the second proposal, although geological evidence indicates the presence of a left-lateral displacement of up to 250 km (Anonymous, 1985) or possibly over 1000 km (Xu et al., 1987) on the Tancheng–Lujiang Fault, most of the motion seems to have taken place during the Indosinian orogeny, i.e. in the Late Triassic (Xu et al., 1987). This motion therefore occurred earlier in time than suggested by the second proposal. Another evidence which does not support large-scale post-Cretaceous displacement along this fault is related to new Cretaceous poles obtained recently from the western region of the Tancheng–Lujiang Fault (Y1, Y2, Y5–Y8, Y10 and S1 in Table 4). These poles are consistent with those obtained on the eastern sector of the fault (Y3, Y4, Y9, S2 and S3 in Table 4), indicating that if such a post-Cretaceous motion indeed exists, the amount of displacement is paleomagnetically insignificant. We therefore conclude that the third proposal is the best interpretation of the present paleomagnetic data for eastern China.

From the above discussion, one must recognize that the Eurasian reference APWP of Besse and Courtillot (1991) probably represents the first path to date which is in agreement with the present understanding of the Mesozoic–Cenozoic tectonic history of Asia. However, it is worth mentioning that a small westward motion of about 8.5° (angular distance) is present between the Early and Late Cretaceous mean poles. As their 95% confidence circles do not overlap, this westward motion is paleomagnetically significant. This phenomenon is not fully concordant with the Eurasian reference APWP of Besse and Courtillot (1991), which shows little displacement in a range of 130–70 Ma, although there is a slight tendency to a westward motion between Early and Late Cretaceous poles (see Fig. 14). Two explanations are plausible. The first is that all the Late Cretaceous poles of eastern China might have a younger age corresponding to the Cretaceous–Tertiary boundary. As can be seen from Fig. 14, our Early Cretaceous mean pole is located at the tip of the hairpin turn, which is in good agreement with the Early Cretaceous APW segment ranging in age from 130 to 110 Ma. However, the Late Cretaceous mean pole is biased towards the APW segment around the Late Cretaceous–Early Tertiary boundary (at approximately 70–60 Ma), and therefore the Late Cretaceous poles for eastern China may correspond to a younger age than those originally reported. The second explanation is that there may be a slight error in the Eurasian reference APWP (Late Cretaceous APW segment) of Besse and Courtillot (1991) caused by transferring poles from other continents onto Eurasia. Enkin et al. (1991) also mentioned this possibility from another phenomenon (latitudinal discrepancy). One argument in favour of this explanation is that the westward motion between Early and Late Cretaceous poles is significant in the Eurasian data alone (about 10° in angular distance; see Table 2(b) of Enkin et al. (1991)) but has been smoothed in the synthetic APWP. As all the Cretaceous poles from various blocks of eastern China, without exception, show this westward motion (see Table 4), it seems unlikely that all the rock units studied happen to have an age around the Cretaceous–

Tertiary boundary. From this point of view, we favour the second explanation unless it is disproved by additional data.

It is noted that this Cretaceous polar motion is different from the proposal of Enkin et al. (1991), which depicted no significant variation of the pole position with time during the Cretaceous. If one re-examines Table 2 of Enkin et al., it is found that mean directions (D_s and I_s in their table) of sites 21–35B are mostly directed in a NNE direction, whereas those of sites 36–43 are mostly directed in a NNW direction. If their site means are divided into two groups, one with sites 21–35B (excluding site 25, as claimed by Enkin et al. (1991)) and another with sites 36–43 (excluding site 42, as claimed by Enkin et al.), a null hypothesis showing that the two groups have a common true mean direction is rejected at a confidence level of 95% using McFadden and Lowes's (1981) method (mean direction of the first group: $D = 15.8°$, $I = 25.2°$, $\kappa = 13.0$, $\alpha_{95} = 8.9°$, $n = 19$ sites; mean direction of the second group: $D = 356.0°$, $I = 31.9°$, $\kappa = 219.3$, $\alpha_{95} = 3.6°$, $n = 7$ sites). As sites 21–35B correspond to an age of 160–110 Ma and sites 36–43 to 85–65 Ma (see Fig. 11 of Enkin et al. (1991)), this rejection in fact indicates a paleomagnetically significant polar motion between the Early and Late Cretaceous. Enkin et al. obtained a different result from ours apparently because of the methodology they used. They applied a least-squares linear fit to the inclinations, declinations and paleolatitudes of all sites separately. However, it seems to be inappropriate to employ this algebraic method for paleomagnetic data because the latter are vectors and not scalar quantities; in this case, a vector analytical method should be applied. Therefore, if our analysis is acceptable, the westward polar motion is also paleomagnetically significant in the result of Enkin et al. (1991).

To summarize, our study favours the proposal that eastern China was fully sutured to Eurasia by the end of the Jurassic or during the Cretaceous and that no paleomagnetically resolvable relative motions have taken place since then. The Cretaceous poles for eastern China display a westward displacement of about 9° (angular distance) from Early to Late Cretaceous. This polar motion is paleomagnetically significant for the Chinese data set but is insignificant for the Eurasian synthetic APWP of Besse and Courtillot (1991). This discrepancy may indicate that all Late Cretaceous poles have a younger age (corresponding to the Cretaceous–Tertiary boundary) than that originally reported, but, more probably it may imply a small error in the Eurasian synthetic APWP. If the latter statement is correct, the Euler poles used for transferring Late Cretaceous paleomagnetic poles from other continents to Eurasia need further revision. However, more Late Mesozoic paleomagnetic studies are needed in other parts of eastern China to confirm our interpretation.

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